

Status of Petrale sole stock off the U.S. West Coast in 2023



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Disclaimer

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1. Executive Summary

1.1. Assessment Model

1.2. Reference Points, Stock Status, and Projections

2. Introduction

Testing adding in an introduction for Petrable sole. There is currently no read of parameters for child documents.

2.1. Stock ID

2.2. Management History

2.3. Fishery Descriptions

2.4. Ecosystem Considerations

Ecosystem considerations and/or climate indicators were not included in this assessment.

3. Data

3.1. Life History

3.2. Catch

3.3. Indices and Standardization

3.4. Composition Data

3.5. Absolute Abundance

3.6. Environmental/Ecosystem Indicator Data

4. Assessment

4.1. Current Modeling Approach

4.2. Configuration of the Base Model

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4.4. Modeling Results

4.4.1. Parameter Estimates

4.4.2. Time Series

4.4.3. Model Fits

4.4.4. Model Diagnostics

4.5. Sensitivity Analyses

4.6. Management Benchmarks

4.7. Projections

5. Discussion

6. Acknowledgements

This document was produced using the R package asar (Schiano, Breitbart, and Saul 2025), which is free to use and publicly available on GitHub.

7. References

Schiano, Samantha, Sophie Breitbart, and Steve Saul. 2025. *Asar: Build NOAA Stock Assessment Report*. <https://github.com/nmfs-ost/asar>.

8. Tables

Table 1: Historical biomass, abundance, and catch time series estimated by the base configuration of the stock assessment model.

Year	Biomass (mt)	Abundance	Catch (mt)
1874	42,648.40	71,606	0.00
1875	42,648.40	71,606	0.00
1876	42,648.70	71,607	2.00
1877	42,647.80	71,606	2.00
1878	42,647.10	71,605	2.00
1879	42,646.60	71,607	2.00
1880	42,646.40	71,609	23.10
1881	42,636.70	71,599	44.20
1882	42,618.30	71,579	65.30
1883	42,591.90	71,550	86.40
1884	42,558.20	71,514	107.50
1885	42,517.80	71,471	128.60
1886	42,471.60	71,422	149.70
1887	42,419.90	71,368	170.80
1888	42,363.50	71,309	191.90
1889	42,302.80	71,246	213.00
1890	42,238.30	71,179	234.10
1891	42,170.30	71,109	255.20
1892	42,099.40	71,037	276.30
1893	42,025.70	70,961	297.42
1894	41,949.60	70,884	318.52
1895	41,871.40	70,804	339.62
1896	41,791.20	70,722	361.20
1897	41,709.20	70,639	382.22
1898	41,625.70	70,555	403.22
1899	41,540.90	70,469	424.32
1900	41,454.90	70,382	445.42
1901	41,367.90	70,294	466.50
1902	41,279.90	70,205	487.60

Year	Biomass (mt)	Abundance	Catch (mt)
1903	41,191.10	70,116	508.68
1904	41,101.50	70,025	529.78
1905	41,011.30	69,934	550.88
1906	40,920.50	69,843	571.96
1907	40,829.20	69,752	593.06
1908	40,737.50	69,660	614.14
1909	40,645.40	69,567	635.24
1910	40,553.00	69,475	656.32
1911	40,460.30	69,382	677.42
1912	40,367.40	69,290	698.52
1913	40,274.30	69,198	719.60
1914	40,181.20	69,106	740.70
1915	40,088.00	69,014	761.78
1916	39,994.70	68,922	773.00
1917	39,906.10	68,836	1,052.98
1918	39,698.50	68,608	847.84
1919	39,603.30	68,523	667.02
1920	39,601.90	68,550	461.10
1921	39,698.80	68,685	587.64
1922	39,733.30	68,733	849.66
1923	39,646.60	68,632	854.82
1924	39,565.00	68,547	1,065.82
1925	39,394.60	68,361	1,057.02
1926	39,243.40	68,208	1,043.42
1927	39,113.40	68,086	1,264.16
1928	38,896.40	67,861	1,240.18
1929	38,711.50	67,683	1,415.16
1930	38,466.30	67,437	1,320.10
1931	38,289.70	67,281	1,282.20
1932	38,153.60	67,179	1,196.78
1933	38,079.00	67,154	957.48
1934	38,131.50	67,284	2,359.90
1935	37,549.60	66,636	1,964.48

Year	Biomass (mt)	Abundance	Catch (mt)
1936	37,203.90	66,327	1,252.62
1937	37,229.20	66,500	2,031.52
1938	36,923.00	66,251	2,247.48
1939	36,543.90	65,882	3,069.56
1940	35,846.20	65,099	2,633.74
1941	35,417.40	64,564	2,353.52
1942	35,141.60	63,999	4,412.12
1943	34,030.70	62,203	4,751.76
1944	32,781.00	59,931	3,242.80
1945	32,118.30	58,135	3,016.52
1946	31,491.40	56,450	5,611.06
1947	29,725.60	53,833	4,445.16
1948	28,492.80	52,408	8,560.90
1949	25,580.20	49,149	6,138.40
1950	23,835.70	47,156	6,922.50
1951	21,938.50	44,975	4,646.88
1952	21,187.10	44,211	4,353.44
1953	20,623.40	43,710	3,856.34
1954	20,284.40	43,454	5,051.98
1955	19,443.80	42,415	4,588.64
1956	18,854.20	41,572	3,698.50
1957	18,675.90	41,223	5,279.48
1958	17,808.60	39,769	4,637.34
1959	17,221.60	38,474	3,761.96
1960	17,042.80	37,801	4,314.92
1961	16,785.30	37,821	5,096.84
1962	16,423.30	40,042	5,381.68
1963	16,206.00	44,246	5,421.12
1964	16,065.30	43,752	4,301.32
1965	16,412.80	42,485	4,218.46
1966	16,826.30	42,090	4,579.02
1967	17,471.20	41,235	4,362.84
1968	18,631.40	58,013	4,242.26

Year	Biomass (mt)	Abundance	Catch (mt)
1969	19,892.30	60,219	4,498.00
1970	20,742.10	54,646	5,427.92
1971	20,690.20	48,048	5,775.36
1972	20,001.20	42,354	5,906.70
1973	18,873.40	38,401	5,901.12
1974	17,414.00	35,729	7,249.66
1975	15,219.80	31,913	6,979.70
1976	13,144.30	28,450	5,528.12
1977	11,806.70	27,072	4,535.88
1978	11,132.40	27,967	6,117.58
1979	9,911.76	29,152	6,109.94
1980	8,660.89	26,363	4,935.20
1981	7,904.04	22,720	4,082.38
1982	7,514.75	21,775	5,260.66
1983	6,618.73	20,820	4,428.38
1984	6,104.36	19,648	3,478.34
1985	6,109.04	19,489	3,678.90
1986	6,178.65	22,436	3,500.86
1987	6,425.57	25,077	4,409.68
1988	6,212.39	23,147	4,298.36
1989	5,912.57	19,371	4,305.40
1990	5,528.16	17,812	3,529.28
1991	5,504.11	19,015	3,854.74
1992	5,420.45	20,372	3,107.00
1993	5,723.00	22,003	3,006.42
1994	6,053.10	21,991	2,750.28
1995	6,533.96	20,626	3,318.10
1996	6,819.65	23,872	3,657.20
1997	6,986.83	23,968	3,895.44
1998	7,039.81	24,249	2,925.54
1999	7,537.44	24,017	2,994.22
2000	8,163.01	24,541	3,786.54
2001	8,608.66	30,679	3,690.44

Year	Biomass (mt)	Abundance	Catch (mt)
2002	9,146.03	31,791	3,594.18
2003	9,589.10	30,448	4,139.16
2004	9,746.01	28,742	3,927.94
2005	9,849.40	27,621	5,468.72
2006	9,201.90	25,441	5,220.40
2007	8,753.74	25,160	4,505.68
2008	9,040.86	27,474	4,439.42
2009	10,012.90	38,274	3,534.88
2010	11,949.10	47,452	1,606.20
2011	15,040.00	58,140	1,869.02
2012	17,973.90	56,835	2,235.44
2013	20,346.80	53,868	4,506.72
2014	21,348.60	50,454	4,817.48
2015	21,799.40	52,737	5,330.34
2016	21,705.90	48,105	5,455.26
2017	21,340.20	47,053	5,861.24
2018	20,599.90	46,072	5,788.88
2019	19,738.60	42,578	5,233.82
2020	18,950.00	39,266	4,184.26
2021	18,500.90	37,665	5,758.22
2022	17,372.50	36,605	6,119.88
2023	16,195.90	35,824	0.00
2024	15,051.60	35,924	0.00

Table 2: Calculated indices of abundance and their corresponding CVs for the fleets and surveys identified in the column headers.

Fleet 3			Fleet 4	
Year	Estimated CPUE	Uncertainty	Estimated CPUE	Uncertainty
1980	3,322.83	0.38		
1983	2,438.46	0.34		
1986	2,211.88	0.34		
1989	2,177.91	0.34		
1992	1,933.47	0.34		

Year	Fleet 3		Fleet 4	
	Estimated CPUE	Uncertainty	Estimated CPUE	Uncertainty
1995	2,516.70	0.35		
1998	2,784.97	0.33		
2001	3,298.33	0.34		
2004	4,051.63	0.34	20,790.6	0.09
2003			19,373.7	0.09
2005			20,698.2	0.08
2006			19,167.6	0.08
2007			17,844.1	0.08
2008			17,074.9	0.08
2009			17,896.0	0.08
2010			22,500.5	0.07
2011			30,760.2	0.07
2012			40,647.6	0.07
2013			49,036.7	0.09
2014			54,277.9	0.07
2015			56,914.9	0.07
2016			57,499.3	0.07
2017			56,575.7	0.07
2018			54,735.8	0.08
2019			52,972.8	0.10
2021			49,724.9	0.07
2022			45,694.1	0.08

9. Figures

Please refer to the `stockplotr` package downloaded from `remotes::install_github('nmfs-ost/stockplotr')` to add premade figures.

A. Appendices

